# PEARL: LCL

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# 1 Introduction

Grid-connected voltage-source inverters require an output filter to attenuate the high-frequency switching harmonics produced by pulse-width modulation before the current is injected into the grid. The LCL filter, consisting of an inverter-side inductor, a shunt capacitor, and a grid-side inductor, has become the standard choice for this role because it provides strong harmonic attenuation with smaller and lighter components than a single inductor of equivalent performance. As a result, the LCL filter is now a near-ubiquitous element of grid-tied inverters in photovoltaic systems, wind turbines and battery energy-storage converters.

Although the LCL filter is a passive element, its three components are subjected to continuous electrical and thermal stress over the service life of the converter. The two inductors carry the full inverter and grid currents and dissipate both core and winding losses, while the capacitor conducts the switching ripple current and is exposed to combined voltage and thermal stress. These losses raise the component temperatures, and elevated temperature is the dominant driver of long-term degradation: the winding-insulation ageing of the inductors follows an Arrhenius-type thermal law, and the film-capacitor wear-out is governed by combined temperature and voltage acceleration. Over the ten-to-twenty-year service life expected of renewable-energy converters, this degradation can make the filter components a limiting factor in overall converter reliability.

## 1.1 Identified research gap

Despite the central role of the LCL filter, reliability assessment in the literature has focused overwhelmingly on the active semiconductor devices and the DC-link capacitor, while the lifetime of the filter components themselves is rarely modelled. Where filter design is treated, the emphasis is almost entirely on the electrical sizing of $L_1$, $L_2$ and $C$ to meet harmonic and resonance constraints, with little attention to the losses, temperatures, and resulting lifetimes of the components under a realistic mission profile. Existing loss and thermal models for magnetic components and film capacitors are typically presented in isolation, and are seldom integrated into a single electro-thermal pipeline that connects the switched electrical waveforms all the way through to a lifetime prediction.

## 1.2 Repository contribution

This documentation presents the LCL-filter lifetime module of PEARL, an open-source electro-thermal reliability framework for power converters. The module:

- reconstructs the full switched electrical behaviour of the LCL filter from the inverter operating point, including the per-harmonic currents and voltages of all three components;
- computes the core and winding losses of the two inductors and the dielectric and ohmic losses of the capacitor;
- estimates the component temperatures through lumped thermal models, and the resulting lifetimes using an Arrhenius ageing law for the inductor insulation and datasheet-based wear-out models for the capacitor;
- evaluates arbitrary mission profiles and accumulates the per-component damage using Miner's rule to obtain an equivalent lifetime and remaining useful life; and
- validates the underlying electrical model against the industry-standard simulator PLECS.

These models are fully integrated with PEARL's electrical and thermal domains, enabling per-sample lifetime estimation of the inverter-side inductor $L_1$, the grid-side inductor $L_2$, and the filter capacitor $C$ under realistic operating profiles.

# 2 Methodology

This section describes the modelling workflow developed to estimate the lifetime of the three LCL-filter components, namely the inverter-side inductor $L_1$, the grid-side inductor $L_2$, and the filter capacitor $C$. The framework follows a physics-informed, multi-domain approach that connects the electrical loading, the component losses, the thermal behaviour, and the lifetime degradation into a single pipeline driven by the inverter mission profile. The full per-function derivations are given in Section 3.4; here only the key governing relations are repeated.

## 2.1 Electrical modelling

The process begins by reconstructing the switched electrical behaviour of the filter for each operating point of the mission profile. From the rated apparent power, grid voltage, and power factor, the required grid current and its phase are determined, and the inverter voltage reference $V_{s,ref}$ needed to drive this current through the filter is computed analytically by phasor analysis of the linear LCL network (Section 3.4.2). The switched inverter voltage $V_s$ is then generated from $V_{s,ref}$ using a volt-second-preserving pulse-width-modulation model (Section 3.4.1), and the filter is solved in the time domain as a discrete state-space system,

$$\mathbf{x}_{k+1} = A_d \mathbf{x}_k + B_d \mathbf{u}_k, \qquad \mathbf{x} = [I_{L_1},\, I_{L_2},\, V_C]^T, \tag{1}$$

yielding the branch voltages and currents of all three components (Section 3.4.3). From these waveforms the RMS values and the per-harmonic spectra used by the loss models are extracted.

## 2.2 Loss modelling

The electrical waveforms are converted into component power losses. For the capacitor, both a dielectric and an ohmic mechanism are evaluated per harmonic and summed (Section 3.4.4):

$$P_{total,C} = \sum_i I(f_i)^2 \left( \frac{\tan \delta_0}{2\pi f_i C} + R_s \right). \tag{2}$$

For each inductor, the loss is split into a core and a winding contribution. The core loss is obtained by applying the Steinmetz equation to every harmonic of the inductor current and summing (Section 3.4.15),

$$P_c = \sum_j k\, f_j^{\alpha}\, B_j^{\beta}\, V_e, \tag{3}$$

while the winding loss is the ohmic dissipation of the RMS current in the DC winding resistance (Section 3.4.16),

$$P_w = R_{dc}\, I_{RMS}^2, \tag{4}$$

so that the total inductor loss is $P_{total} = P_c + P_w$.

## 2.3 Thermal modelling

Each component temperature is obtained from a single-node steady-state thermal model, in which the temperature rise above ambient is proportional to the dissipated power through a lumped thermal resistance:

$$T = T_{amb} + R_{th} P_{total}. \tag{5}$$

The thermal resistance of each component is estimated from its exposed surface area and the convective heat-transfer coefficient (Sections 3.4.6 and 3.4.17). The thermal capacitance is neglected, which is justified for mission-profile analysis where each operating point persists long relative to the component thermal time constants.

## 2.4 Lifetime modelling

The instantaneous component temperatures and voltages are mapped to a predicted lifetime. For the inductor winding insulation, an Arrhenius thermal-ageing law is used, in which the lifetime relative to its rated value depends on the difference between the rated and operating temperatures through the activation energy of the insulation. For the capacitor, the lifetime is obtained either analytically, from combined voltage and temperature acceleration, or graphically, by interpolation of the datasheet lifetime curves, according to the selected method. Each model returns a predicted time-to-failure $L_i$ corresponding to the operating conditions at mission-profile sample $i$.

## 2.5 Cumulative damage assessment (Miner's rule)

For the time-varying mission profile, the lifetime consumption of each component is accumulated using Miner's linear damage rule. Given the wall-clock duration $\Delta t$ represented by each sample, the damage increment of sample $i$ is

$$d_i = \frac{\Delta t}{L_i}, \qquad D = \sum_i d_i, \tag{6}$$

and the equivalent lifetime of a periodically repeating mission profile is

$$L_{eq} = \frac{T_{profile}}{D}, \tag{7}$$

where $T_{profile}$ is the total duration of one mission profile. The result is reported in years for each component, together with the fraction of lifetime consumed.

## 2.6 Constraint and consistency checking

Throughout the workflow, the framework enforces a set of physical and datasheet limits to ensure that every evaluated operating point is valid. These include the achievable inverter voltage, the core saturation flux density and winding-window fill of the inductors, the capacitor voltage, current and temperature ratings, and the agreement between the chosen filter components and an independent LCL design (Section 3.2). A violation of any constraint raises an error and terminates the run, so that the lifetime results are produced only for physically consistent designs.

# 3 Detailed description of python files

## 3.1 Input_parameters

This module initializes and stores all input parameters required by the LCL-filter lifetime model. It defines the system-level inverter ratings, the PWM and time-discretization settings, the mission profile, the ambient-temperature profile, the LCL-filter design parameters, and the complete physical, thermal and lifetime specifications of the three filter components, namely the inverter-side inductor $L_1$, the grid-side inductor $L_2$, and the filter capacitor $C$. All quantities are centralized within a single class, `Input_parameters_class`, which serves as the single source of truth from which every downstream computation, validation, and plotting routine draws its values.

### 3.1.1 System parameters

This block defines the top-level inverter and grid operating parameters that govern the electrical model. These quantities set the DC-bus voltage, the inverter topology and modulation strategy, and the fundamental and switching frequencies from which all time-discretization and waveform-generation steps are derived. Table 1 summarizes these parameters.

Table 1: System-level inverter and grid configuration parameters.

| **Variable** | **Type** | **Unit** | **Description** |
|---|---|---|---|
| `Vdc_rated` | float | V | Rated DC-bus voltage. |
| `Vo_rated` | float | V | Rated PWM pulse amplitude ($\pm V_o$). |
| `inverter_phases` | int | – | Number of phases: 1 = single-phase, 3 = three-phase. |
| `M_rated` | float | – | Modulation index ($0 \leq M \leq 1$ in the linear region). |
| `single_phase_inverter_topology` | str | – | Single-phase topology: `"half"` = $\pm V_{dc}/2$, `"full"` = $\pm V_{dc}$. |
| `waveform_voltage_definition` | str | – | Voltage definition: `"switched_output"` or `"pole_voltage"` ($\pm V_{dc}/2$). |
| `modulation_scheme` | str | – | PWM strategy: `"spwm"` or `"svm"`. Only `"spwm"` is supported. |
| `f` | float | Hz | Fundamental frequency. |
| `fsw` | float | Hz | Switching (carrier) frequency. |
| `T` | float | s | Fundamental period. |
| `Tsw` | float | s | Switching period. |
| `omega` | float | rad/s | Fundamental angular frequency. |

### 3.1.2 Mission profile

The mission profile defines the time-varying operating conditions applied to the inverter over the simulated service period. Each array entry represents one operating point sampled at one-second resolution, from which the active power, reactive power, and required grid-side current are derived. Table 2 summarizes these parameters.

Table 2: Mission-profile parameters.

| Variable | Type | Unit | Description |
|---|---|---|---|
| `Profile_size` | int | – | Number of operating points in the mission profile. |
| `Vdc_RMS` | array | V | DC-bus voltage profile. |
| `M` | array | – | Modulation-index profile. |
| `Vo` | array | V | PWM pulse-amplitude profile. |
| `Vg_RMS` | array | V | Grid-side RMS AC voltage profile. |
| `S_RMS` | array | VA | Apparent-power profile. |
| `pf` | array | – | Power-factor profile; positive = capacitive, negative = inductive. |
| `P_RMS` | array | W | Active power. |
| `Q_RMS` | array | VAr | Reactive power. |
| `Ig_RMS` | array | A | Required grid-side current. |

### 3.1.3 Thermal parameters

These parameters define the thermal boundary conditions applied to the filter components. The ambient-temperature profile sets the reference temperature for each operating point and the heat-transfer coefficient governs the convective cooling used in the component thermal resistance calculations. Table 3 summarizes these parameters.

Table 3: Thermal parameters.

| Variable | Type | Unit | Description |
|---|---|---|---|
| `T_amb` | array | K | Ambient-temperature profile. |
| `heat_transfer_coefficient` | float | $W/(m^2 \cdot K)$ | Convective heat-transfer coefficient for cooling. |

### 3.1.4 Time discretization and simulation resolution

These parameters define the temporal discretization of the simulation. The number of samples per fundamental cycle sets the waveform fidelity and the derived time step, while the samples per switching period check ensures the PWM switching events are adequately resolved. The wall-clock duration per sample maps each mission-profile point onto real elapsed time for the lifetime accumulation. Table 4 summarizes these parameters.

Table 4: Time-discretization and simulation-resolution parameters.

| Variable | Type | Unit | Description |
|---|---|---|---|
| `resolution_per_cycle` | int | – | Number of samples representing one fundamental AC cycle. |
| `dt` | float | s | Simulation time step. |
| `samples_per_switching_period` | float | – | Number of samples within one PWM switching period. |
| `Minimum_required_samples_per_switching_period` | int | – | Minimum samples per switching period required to resolve switching events. |
| `seconds_per_sample` | int | s | Wall-clock duration represented by each mission-profile sample (e.g. 86400 = one day). |

### 3.1.5 LCL filter design parameters

These parameters define the inputs to the LCL-filter sizing procedure. They specify the rated electrical operating point of the inverter and the two design constraints, namely the allowable inverter-side current ripple and the harmonic attenuation ratio, from which the filter components $L_1$, $L_2$, $C$, and the damping resistor $R_3$ are derived. Table 5 summarizes these parameters.

Table 5: LCL-filter design parameters.

| **Variable** | **Type** | **Unit** | **Description** |
|---|---|---|---|
| `Vg_ll_RMS` | float | V | RMS of the fundamental line-to-line grid voltage. |
| `S_rated` | float | VA | Rated apparent power of the inverter. |
| `I_rated_RMS` | float | A | Rated RMS current, $I = S/(\sqrt{3}\,V_{g,ll})$. |
| `I_rated_peak` | float | A | Peak rated current, $\sqrt{2}\,I_{RMS}$. |
| `current_ripple_limit` | float | – | Allowed inverter-side current ripple as a fraction of rated current. |
| `delta` | float | – | Harmonic attenuation ratio used in the LCL design. |
| `omega_sw` | float | rad/s | Switching angular frequency, $\omega_{sw} = 2\pi f_{sw}$. |

### 3.1.6 Inverter-side inductor properties

This dictionary defines the complete specification of the inverter-side inductor $L_1$. It groups the inductance value, the core material and Steinmetz loss coefficients, the core geometry, the copper winding parameters, and the insulation lifetime constants used in the Arrhenius ageing model. Table 6 summarizes these parameters.

Table 6: Inverter-side inductor ($L_1$) specifications.

| Variable | Type | Unit | Description |
|---|---|---|---|
| L1 | float | H | Inductance. |
| k | float | W/m$^3$ | Steinmetz coefficient $k$ (core-loss scale). |
| a | float | – | Steinmetz frequency exponent $\alpha$. |
| b | float | – | Steinmetz flux-density exponent $\beta$. |
| rho_mass | float | kg/m$^3$ | Core material density. |
| Bsat | float | T | Core saturation flux density. |
| B_max | float | T | Operating flux-density limit (70% of $B_{sat}$). |
| mu_r | float | – | Relative permeability (assumed constant). |
| A_core | float | m | Overall core width. |
| B_core | float | m | Overall core height. |
| D_core | float | m | Core depth (cast width). |
| E_core | float | m | Core thickness (build). |
| F_core | float | m | Window width (winding opening). |
| G_core | float | m | Window height (winding opening). |
| kf | float | – | Core stacking factor. |
| J_max | float | A/m$^2$ | Maximum winding current density. |
| d_strand | float | m | Strand diameter. |
| A_strand | float | m$^2$ | Bare copper area of one strand. |
| rho | float | Ω·m | Material resistivity (assumed constant). |
| T_insulation_rated | float | K | Rated insulation temperature index. |
| L_insulation_rated | float | h | Reference insulation lifetime at the rated temperature. |
| Ea_insulation | float | J | Activation energy of the insulation ageing mechanism. |
| kb | float | J/K | Boltzmann constant. |
| V_bd | float | V | Minimum winding breakdown voltage. |
| mu_0 | float | H/m | Permeability of free space. |
| R1 | float | Ω | Inductor series resistance. |
| L_max_years | float | years | Maximum lifetime cap. |

### 3.1.7 Grid-side inductor properties

This dictionary defines the complete specification of the grid-side inductor $L_2$. It follows the same structure as the inverter-side inductor, grouping the inductance value, the core material and Steinmetz loss coefficients, the core geometry, the copper winding parameters, and the insulation lifetime constants. Table 7 summarizes these parameters.

Table 7: Inverter-side inductor ($L_2$) specifications.

| Variable | Type | Unit | Description |
|---|---|---|---|
| L2 | float | H | Inductance. |
| k | float | W/m$^3$ | Steinmetz coefficient $k$ (core-loss scale). |
| a | float | – | Steinmetz frequency exponent $\alpha$. |
| b | float | – | Steinmetz flux-density exponent $\beta$. |
| rho_mass | float | kg/m$^3$ | Core material density. |
| Bsat | float | T | Core saturation flux density. |
| B_max | float | T | Operating flux-density limit (70% of $B_{sat}$). |
| mu_r | float | – | Relative permeability (assumed constant). |
| A_core | float | m | Overall core width. |
| B_core | float | m | Overall core height. |
| D_core | float | m | Core depth (cast width). |
| E_core | float | m | Core thickness (build). |
| F_core | float | m | Window width (winding opening). |
| G_core | float | m | Window height (winding opening). |
| kf | float | – | Core stacking factor. |
| J_max | float | A/m$^2$ | Maximum winding current density. |
| d_strand | float | m | Strand diameter. |
| A_strand | float | m$^2$ | Bare copper area of one strand. |
| rho | float | Ω·m | Material resistivity (assumed constant). |
| T_insulation_rated | float | K | Rated insulation temperature index. |
| L_insulation_rated | float | h | Reference insulation lifetime at the rated temperature. |
| Ea_insulation | float | J | Activation energy of the insulation ageing mechanism. |
| kb | float | J/K | Boltzmann constant. |
| V_bd | float | V | Minimum winding breakdown voltage. |
| mu_0 | float | H/m | Permeability of free space. |
| R2 | float | Ω | Inductor series resistance. |
| L_max_years | float | years | Maximum lifetime cap. |

### 3.1.8 Filter capacitor properties

This dictionary defines the complete specification of the LCL-filter capacitor $C$ together with its series damping resistor $R_3$. It groups the capacitance value, the case dimensions, the electrical and thermal ratings, the dissipation-factor loss-model parameters, the datasheet lifetime curves, and the selected lifetime-calculation method. Some entries are initialized to `None` and are computed at runtime from the other parameters. Table 8 summarizes these parameters.

Table 8: Filter capacitor ($C$) specifications.

| Variable | Type | Unit | Description |
|---|---|---|---|
| `C` | float | F | Capacitance. |
| `D_case` | float | m | Case diameter. |
| `H_case` | float | m | Case height. |
| `I_C_RMS_rated` | float | A | Rated RMS current. |
| `V_C_RMS_Rated` | float | V | Rated RMS voltage. |
| `V_C_Peak_Rated` | float | V | Rated peak voltage. |
| `Temperature_Rated` | float | K | Rated temperature. |
| `Lifetime_Rated` | float | h | Rated lifetime. |
| `A` | float | – | Voltage-acceleration constant. |
| `n` | float | – | Voltage-acceleration exponent. |
| `T_C_Rated` | float | K | Maximum hotspot temperature. |
| `Thermal_resistance_C` | float | K/W | Core-to-ambient thermal resistance. |
| `Rs` | float | Ω | Equivalent series resistance (ESR). |
| `tan_delta_measured` | float | – | Measured dissipation factor. |
| `f_measured_for_tan_delta` | float | Hz | Dissipation factor at measured frequency. |
| `tan_delta_0` | float | – | Dielectric loss tangent; computed at runtime if `None`. |
| `lifetime_curves` | dict | K, h | Datasheet lifetime curves ($L$ vs. $T$) at each voltage stress ratio $V/V_r$. |
| `Lifetime_calculations` | str | – | Lifetime method: `"Graphical"` or `"Analytical"`. |
| `R3` | float | Ω | Series damping resistor. |

## 3.2 LCL_filter_design

This module implements the sizing procedure for the passive components of the LCL filter used in the grid-connected inverter. The function `LCL_filter_design_function` determines the inverter-side inductance $L_1$, grid-side inductance $L_2$, filter capacitance $C$, and passive damping resistor $R_3$ from the rated operating point and two design constraints, namely the allowable inverter-side current ripple and the harmonic attenuation ratio. The procedure selects a candidate design that simultaneously satisfies the maximum-capacitance, total-inductance, and resonance-frequency constraints, and raises detailed diagnostic errors when no feasible design exists. The methodology follows Han et al. [1].

### 3.2.1 Capacitance limit

The filter capacitor exchanges reactive power with the grid, which slightly reduces the converter power factor. To keep this exchange small, the capacitance is limited so that the reactive power absorbed by $C$ does not exceed 5% of the rated power:

$$C_{\max} = \frac{0.05\, S_{\text{rated}}}{2\pi f_o\, V_{g,ll}^2}, \tag{8}$$

where $S_{\text{rated}}$ is the rated apparent power, $f_o$ the fundamental frequency, and $V_{g,ll}$ the line-to-line grid voltage.

### 3.2.2 Total inductance limit

A large total inductance increases the fundamental voltage drop across the filter and the required DC-link voltage. The combined inductance is therefore constrained so that the voltage drop does not exceed 10%:

$$L_{T,\max} = \frac{0.10\,V_{g,ll}^2}{2\pi f_o\, S_{\text{rated}}}, \qquad L_1 + L_2 \le L_{T,\max}. \tag{9}$$

### 3.2.3 Inverter-side inductance

The inverter-side inductor $L_1$ limits the high-frequency switching-current ripple. Its minimum value is set by the allowed peak-to-peak ripple as a fraction of the rated current. For a three-phase inverter,

$$L_{1,\min} = \frac{\sqrt{3}}{12}\,\frac{U_{dc}\,M}{\Delta i_L\, I_{\text{rated}}\, f_{sw}}, \tag{10}$$

where $U_{dc}$ is the DC-link voltage, $M$ the modulation index, $f_{sw}$ the switching frequency, and $\Delta i_L$ the current-ripple limit. For a single-phase bipolar SPWM inverter the corresponding expression is $L_{1,\min} = U_{dc}/(\Delta i_L\, I_{\text{rated}}\, r\, f_{sw})$ with $r = 2$.

### 3.2.4 Grid-side inductance and resonance

The grid-side inductor $L_2$ is obtained from the harmonic attenuation ratio $\delta$, which sets how strongly the switching-frequency current ripple is attenuated before reaching the grid. Writing $L_2 = a_L L_1$,

$$a_L = \left|\frac{(1/\delta) - 1}{1 - L_1 C\,\omega_{sw}^2}\right|, \tag{11}$$

where $\omega_{sw} = 2\pi f_{sw}$. For each candidate capacitance the resonance frequency of the filter is

$$f_r = \frac{1}{2\pi}\sqrt{\frac{L_1 + L_2}{L_1 L_2 C}}. \tag{12}$$

To avoid amplifying low-order harmonics or interfering with the switching band, the resonance frequency must satisfy

$$10 f_o < f_r < 0.5 f_{sw}. \tag{13}$$

The function sweeps a range of capacitance values, retains those satisfying the total-inductance and resonance constraints, and selects the candidate whose capacitance is closest to $C_{\max}/2$.

### 3.2.5 Passive damping resistor

To suppress the LCL resonance peak, a damping resistor $R_3$ is placed in series with the filter capacitor (the PD-3 method). Its value is chosen as the geometric mean of a lower bound, ensuring sufficient damping, and an upper bound, limiting the added losses:

$$R_{3,\min} = \frac{1}{6\pi}\,\frac{L_2 f_{sw}}{L_1 f_r}\,\frac{1}{C\,\omega_r}, \qquad R_{3,\max} = \frac{1}{\omega_{sw} C}, \qquad R_3 = \sqrt{R_{3,\min}\, R_{3,\max}}, \tag{14}$$

where $\omega_r = 2\pi f_r$.

## 3.3 mother_function

The `mother_function` module is the top-level executable of the LCL-filter lifetime framework. It orchestrates the complete workflow by loading the input parameters, validating them, designing and cross-checking the filter, computing the operating-point-independent component constants once, and then evaluating each mission-profile sample to obtain the electrical, loss, thermal and lifetime quantities of the three filter components. The per-component lifetimes are finally aggregated over the mission profile using Miner's rule, and the results are stored and visualized. The workflow is summarized in Algorithm 1.

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**Algorithm 1:** LCL-filter component lifetime estimation workflow.

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**Input:** Mission profile $(P, Q, V_{dc}, V_g, S, pf, T_{amb})$; inverter and PWM parameters; LCL design parameters; component specifications ($L_1$, $L_2$, $C$).

**Output:** Equivalent lifetime and lifetime consumed for $C$, $L_1$, $L_2$; result dataframes; figures.

Validate all input parameters;

Design reference filter and cross-check chosen $(L_1, L_2, C, R_3)$;

Compute operating-point-independent constants for $C$, $L_1$, $L_2$;

**foreach** *mission-profile sample i* **do**

- Build $V_{s,ref}$ and generate switched voltage $V_s$;
- Solve LCL filter for $V_{L_1}, I_{L_1}, V_C, I_C, V_{L_2}, I_{L_2}$;
- Compute RMS values, losses and temperatures of all components;

Concatenate per-sample results into continuous time series;

Compute instantaneous lifetimes ($C$: graphical/analytical; $L_1, L_2$: Arrhenius);

Apply Miner's rule to obtain equivalent lifetime and lifetime consumed;

Store results and generate plots;

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**Input validation.**

Before any computation, a series of validation routines from `Calculation_functions_class` verify the consistency of the inputs: the achievable AC RMS voltage, the PWM pulse amplitude, the simulation resolution, the agreement between phase and line-to-line grid voltage, the mission-profile array lengths, the power-factor range, and the apparent-power limit. Any violation raises an error and terminates the run.

**Filter design cross-check.**

The independent sizing routine `LCL_filter_design_function` (Section 3.2) is called to compute an optimal reference set $(L_1, L_2, C, R_3)$. The user-specified component values are then checked against this reference using `check_within_tolerance`, confirming that the chosen components lie within an acceptable tolerance of a valid design.

**Component derived constants.**

Quantities that do not depend on the operating point are computed once for each component. For the capacitor, the thermal resistance and dielectric loss tangent $\tan\delta_0$ are derived from the case geometry and datasheet dissipation factor. For each inductor, the core surface area, effective cross-sectional area $A_e$, magnetic path length $l_e$, core volume $V_e$, number of turns $N$, air-gap length $l_g$, peak flux density, winding cross-section and parallel-strand count, mean turn length, DC winding resistance $R_{dc}$, and thermal resistance $R_{th}$ are computed, accompanied by saturation and window-fill safety checks.

**Per-setpoint solver.**

The core electrical evaluation is performed by `solve_setpoint`, a memoized function (`lru_cache`) that solves one mission-profile operating point. For a given setpoint it constructs the reference inverter voltage $V_{s,ref}$, generates the switched PWM voltage $V_s$, solves the LCL filter to obtain the branch voltages and currents $(V_{L_1}, I_{L_1}, V_C, I_C, V_{L_2}, I_{L_2})$, and from these computes the RMS quantities, per-harmonic losses, total losses and core/hotspot temperatures of all three components. Caching avoids recomputation for repeated operating points in the mission profile.

**Mission-profile evaluation.**

The solver is applied to every sample of the mission profile, and the resulting per-sample arrays are concatenated into continuous time series for each electrical, loss and thermal quantity.

**Lifetime aggregation.**

For the capacitor, the instantaneous lifetime is obtained either graphically (datasheet curves) or analytically, according to the selected method. For each inductor, the instantaneous winding-insulation lifetime is obtained from the Arrhenius ageing model. In all three cases, the per-second lifetimes are accumulated over the mission profile using the modified Miner's rule (`miners_rule_modified`), yielding the equivalent lifetime and the fraction of lifetime consumed.

**Reporting and visualization.**

All electrical, thermal and lifetime results are consolidated into per-component report dictionaries and dataframes, which are passed to the plotting module (Section 3.5) to generate the mission-profile, waveform and per-component figures.

### 3.3.1 LCL design cross-check variables

These variables hold the reference component values returned by the independent sizing routine `LCL_filter_design_function`, against which the user-specified components are validated. Table 9 summarizes these variables.

Table 9: LCL design cross-check variables.

| Variable | Type | Unit | Description |
|---|---|---|---|
| `L1_optimum` | float | H | Reference inverter-side inductance from the design routine. |
| `L2_optimum` | float | H | Reference grid-side inductance from the design routine. |
| `C_optimum` | float | F | Reference filter capacitance from the design routine. |
| `R3_optimum` | float | Ω | Reference damping resistance from the design routine. |

### 3.3.2 Inductor derived constants

These variables are the operating-point-independent geometric, magnetic, winding and thermal constants computed once for each inductor ($L_1$ and $L_2$) from its core and winding specifications. The same set is computed for both inductors; the suffix `_L1` or `_L2` identifies the component. Table 10 summarizes these variables.

Table 10: Per-inductor derived constants (suffix _L1 / _L2).

| Variable | Type | Unit | Description |
|---|---|---|---|
| A_surface | float | $m^2$ | Core outer surface area used for thermal-resistance estimation. |
| Ae | float | $m^2$ | Effective core cross-sectional area. |
| le | float | m | Effective magnetic path length. |
| Ve | float | $m^3$ | Effective core volume. |
| N | int | – | Number of winding turns. |
| lg | float | m | Air-gap length. |
| B_peak | float | T | Peak core flux density at rated current. |
| A_wire_minimum | float | $m^2$ | Minimum copper cross-section required for the rated current. |
| N_parallel_ wire | int | – | Number of parallel strands used. |
| A_wire_actual | float | $m^2$ | Actual copper cross-section after rounding up strands. |
| l_turn | float | m | Mean length of one winding turn. |
| Rdc | float | Ω | DC winding resistance (skin and proximity effects neglected). |
| R_th | float | K/W | Core-to-ambient thermal resistance. |

### 3.3.3 Per-operating-point solver outputs

These variables are produced by `solve_setpoint` for each mission-profile sample and, after the mission-profile loop, concatenated into continuous time series. They comprise the instantaneous waveforms, the per-component RMS quantities, losses, and temperatures. Table 11 summarizes these variables.

Table 11: Per-operating-point solver output variables.

| Variable | Type | Unit | Description |
|---|---|---|---|
| Vg | array | V | Instantaneous grid voltage waveform. |
| pf_inst | array | – | Instantaneous power factor. |
| phi | array | rad | Phase-angle magnitude, $\arccos|pf|$. |
| phase_shift | array | rad | Signed phase shift applied to the current reference. |
| Ig_ref | array | A | Reference grid current waveform. |
| Vs_ref | array | V | Reference inverter voltage waveform. |
| Vs | array | V | Switched PWM inverter output voltage. |
| V_L1, I_L1 | array | V, A | Inverter-side inductor voltage and current waveforms. |
| V_C, I_C | array | V, A | Capacitor voltage and current waveforms. |
| V_L2, I_L2 | array | V, A | Grid-side inductor voltage and current waveforms. |
| I_C_RMS_harmonics | array | A | Per-harmonic RMS capacitor current. |
| I_C_RMS | float | A | Total RMS capacitor current. |
| V_C_RMS | float | V | RMS capacitor voltage. |
| P_total_C | float | W | Total capacitor power loss. |
| T_C | float | K | Capacitor hotspot temperature. |
| V_L1_RMS, I_L1_RMS | float | V, A | RMS voltage and current of the inverter-side inductor. |
| P_c_L1, P_w_L1 | float | W | Core and winding losses of the inverter-side inductor. |
| P_total_L1 | float | W | Total loss of the inverter-side inductor. |
| T_inductor_L1 | float | K | Inverter-side inductor temperature. |
| V_L2_RMS, I_L2_RMS | float | V, A | RMS voltage and current of the grid-side inductor. |
| P_c_L2, P_w_L2 | float | W | Core and winding losses of the grid-side inductor. |
| P_total_L2 | float | W | Total loss of the grid-side inductor. |
| T_inductor_L2 | float | K | Grid-side inductor temperature. |

#### 3.3.4 Lifetime and THD variables

These variables hold the harmonic-distortion metrics and the per-component lifetime results obtained after the mission-profile evaluation. The lifetime of each component is computed as a series and then aggregated using the modified Miner's rule. Table 12 summarizes these variables.

Table 12: Lifetime and THD variables.

| Variable | Type | Unit | Description |
|---|---|---|---|
| THD_percent_ I_L2 | float | % | Total harmonic distortion of the grid-side current $I_{L_2}$. |
| THD_percent_Vs | float | % | Total harmonic distortion of the switched inverter voltage $V_s$. |
| Lifetime_C_series | array | years | Predicted lifetime of the capacitor over the mission profile. |
| Lifetime_L1_series | array | years | Predicted lifetime of the inverter-side inductor over the mission profile. |
| Lifetime_L2_series | array | years | Predicted lifetime of the grid-side inductor over the mission profile. |
| Lifetime_C | float | years | Equivalent lifetime of the capacitor after Miner's rule. |
| Lifetime_L1 | float | years | Equivalent lifetime of the inverter-side inductor after Miner's rule. |
| Lifetime_L2 | float | years | Equivalent lifetime of the grid-side inductor after Miner's rule. |
| Lifetime_consumed_C | float | % | Fraction of capacitor lifetime consumed over one mission profile. |
| Lifetime_consumed_L1 | float | % | Fraction of inverter-side inductor lifetime consumed over one mission profile. |
| Lifetime_consumed_L2 | float | % | Fraction of grid-side inductor lifetime consumed over one mission profile. |

### 3.3.5 Output dataframes

The final results are organized into five pandas dataframes, each grouping a related set of the previously defined quantities for storage and visualization. Each dataframe is optionally exported to Parquet and passed to the corresponding plotting routine. Table 13 summarizes these dataframes.

Table 13: Output dataframes.

| Dataframe | Type | Description |
|---|---|---|
| df_1_power_flow_RMS | DataFrame | RMS mission-profile quantities: $V_{dc}$, $V_g$, $S$, $pf$, $P$, $Q$, and $I_g$, one row per mission-profile sample. |
| df_2_power_flow_inst | DataFrame | Instantaneous one-cycle waveforms: $pf$, $\phi$, $I_{g,ref}$, $V_{s,ref}$, $V_s$, and the per-component voltage/current pairs, together with the THD of $I_{L_2}$ and $V_s$ stored on the final row. |
| df_3_C | DataFrame | Capacitor results: RMS voltage and current, total loss, hotspot temperature, equivalent lifetime and lifetime consumed. |
| df_4_L1 | DataFrame | Inverter-side inductor results: per-sample RMS quantities, losses and temperature, plus the scalar geometry, winding and lifetime constants stored on the final row. |
| df_5_L2 | DataFrame | Grid-side inductor results: identical structure to df_4_L1 for the grid-side inductor. |

## 3.4 Calculation_functions

The `Calculation_functions` module is the computational core of the LCL-filter lifetime framework. It is implemented as a single class, `Calculation_functions_class`, containing a structured collection of static methods that implement the analytical and numerical equations governing the electrical model, the component losses, the thermal behaviour, and the lifetime estimation of the three filter components. The module also provides an extensive set of input-validation and safety-check routines that verify the physical consistency of the inputs and enforce the operating limits of each component throughout the simulation.
Every function in this module is documented in detail within the source code through descriptive names, in-line comments, and docstrings that state its purpose, inputs, outputs and the references on which it is based. For this reason, an exhaustive description of every function is not repeated here. Instead, the following subsections describe only those functions for which the underlying physics warrants additional explanation, as the author considers their theoretical background important for understanding the model. All remaining functions are considered sufficiently self-explanatory from the source code itself.

### 3.4.1 Three_phase_switching_output

This function generates the switched phase voltage $V_s$ produced by a three-phase, three-wire inverter from the sinusoidal reference voltage $V_{s,ref}$. Rather than comparing the reference against a sampled carrier point by point, it computes the exact switching instants analytically and renders them onto the simulation time grid using volt-second-preserving averaging. This produces a switched waveform whose local average is exact regardless of the grid resolution.

**Three-phase references.**

Three modulating references are constructed at 0°, −120° and +120° by time-shifting $V_{s,ref}$ by one third of the fundamental period. Each is normalized by the pole-voltage amplitude $V_o = V_{dc}/2$ to obtain the modulating signals $m_a, m_b, m_c$.

**Zero-sequence injection.**

A min–max zero-sequence component is added to all three modulating signals to implement space-vector modulation, which extends the linear modulation range:

$$v_{zs} = -\frac{1}{2}\left(\max(m_a, m_b, m_c) + \min(m_a, m_b, m_c)\right), \tag{15}$$

after which each signal is offset by $v_{zs}$ and clipped to $[-1, 1]$.

**Analytic switching instants.**

The carrier is a symmetric triangular wave normalized over each switching period, $c(\tau) = 4\,|\tau - 0.5| - 1$ with $\tau = (t \bmod T_{sw})/T_{sw}$. A leg is on ($+V_o$) when its modulating signal exceeds the carrier, $m \geq c$, which occurs over the interval $[\tau_1, \tau_2]$ with

$$\tau_1 = \frac{1-m}{4}, \qquad \tau_2 = \frac{m+3}{4}, \qquad d = \tau_2 - \tau_1 = \frac{m+1}{2}, \tag{16}$$

where $d$ is the per-period duty ratio. The modulating signal is sampled once per carrier period (symmetric regular sampling), so the rising and falling edges are placed exactly from the carrier geometry, independent of the simulation grid.

**Volt-second-preserving rendering.**

Each leg is rendered onto the coarse time grid by assigning every sample the *time-average* of the pole voltage over its $\Delta t$ window, rather than an instantaneous point sample. The cumulative on-time function is evaluated at the cell boundaries and differenced to obtain the on-time of each cell; the on fraction is then $f_{on} = t_{on}/\Delta t$, and the averaged pole voltage of each leg is

$$V_{leg} = V_o\,(2f_{on} - 1), \tag{17}$$

which equals $+V_o$ for a fully-on cell and $-V_o$ for a fully-off cell. This averaging conserves the volt-seconds of each pulse and is inherently anti-aliased, avoiding the edge jitter that point-sampling onto a coarse grid would introduce.

**Common-mode removal.**

Finally, the phase voltage seen by the three-wire load is obtained by removing the common-mode component from the three pole voltages:

$$V_s = V_{ao} - \frac{V_{ao} + V_{bo} + V_{co}}{3}. \tag{18}$$

### 3.4.2 compute_Vs_ref_phasor

This function computes the inverter voltage reference $V_{s,ref}$ required to drive a specified grid current into the grid through the LCL filter. Because the reference grid current $I_{g,ref}$ and the grid voltage $V_g$ are pure sinusoids at the fundamental frequency, and the LCL filter is a linear network, the problem can be solved exactly in the frequency domain using phasor analysis. This avoids any numerical integration error in establishing the reference, in contrast to a time-domain solution.

**Branch impedances.**

At the fundamental angular frequency $\omega = 2\pi f$, the three branches of the filter are represented by their complex impedances:

$$Z_{L_1} = R_1 + j\omega L_1, \qquad Z_{L_2} = R_2 + j\omega L_2, \qquad Z_C = R_3 + \frac{1}{j\omega C}, \tag{19}$$

where $Z_C$ includes the series damping resistance $R_3$ in the capacitor branch.

**Phasor solution.**

The known quantities are the grid-current phasor and the grid-voltage phasor. Taking the grid voltage as the phase reference,

$$\bar{I}_{L_2} = \sqrt{2}\, I_{g,RMS}\, e^{j\varphi}, \qquad \bar{V}_g = \sqrt{2}\, V_{g,RMS}\, e^{j0}, \tag{20}$$

where $\varphi$ is the phase shift between current and voltage. The network is then solved branch by branch. The voltage at the node between the two inductors follows from Kirchhoff's voltage law on the grid side:

$$\bar{V}_{node} = \bar{V}_g + Z_{L_2}\,\bar{I}_{L_2}. \tag{21}$$

The capacitor-branch current is obtained from this node voltage,

$$\bar{I}_C = \frac{\bar{V}_{node}}{Z_C}, \tag{22}$$

and Kirchhoff's current law gives the inverter-side inductor current, from which the required inverter voltage follows by Kirchhoff's voltage law:

$$\bar{I}_{L_1} = \bar{I}_{L_2} + \bar{I}_C, \qquad \bar{V}_s = \bar{V}_{node} + Z_{L_1}\,\bar{I}_{L_1}. \tag{23}$$

**Time-domain reconstruction.**

The resulting inverter-voltage phasor $\bar{V}_s$ is converted back to a time-domain waveform for each mission-profile segment using its magnitude and angle:

$$V_{s,ref}(t) = |\bar{V}_s| \sin\big(\omega t + \angle\bar{V}_s\big). \tag{24}$$

The computation is repeated for each second of the mission profile, since the current amplitude, voltage amplitude and phase shift may differ between operating points.

### 3.4.3 LCL_Filter_Grid_Connected

This function solves the time-domain response of the LCL filter connected between the inverter and the grid. Given the switched inverter voltage $V_s$ and the grid voltage $V_g$, it computes the two inductor currents, the capacitor voltage, and the derived branch voltages and capacitor current. The filter is described by three state variables, namely the inverter-side inductor current $I_{L_1}$, the grid-side inductor current $I_{L_2}$, and the capacitor voltage $V_C$.

**Governing equations.**

Applying Kirchhoff's voltage and current laws to the LCL network with the parasitic series resistances $R_1$, $R_2$ and the capacitor-branch damping resistance $R_3$ yields the following set of coupled first-order differential equations:

$$\frac{dI_{L_1}}{dt} = \frac{V_s - R_1 I_{L_1} - V_C - R_3(I_{L_1} - I_{L_2})}{L_1}, \tag{25}$$

$$\frac{dI_{L_2}}{dt} = \frac{V_C + R_3(I_{L_1} - I_{L_2}) - V_g - R_2 I_{L_2}}{L_2}, \tag{26}$$

$$\frac{dV_C}{dt} = \frac{I_{L_1} - I_{L_2}}{C}. \tag{27}$$

**State-space formulation.**

Collecting the state variables into the vector $\mathbf{x} = [I_{L_1}, I_{L_2}, V_C]^T$ and the inputs into $\mathbf{u} = [V_s, V_g]^T$, the system is written in the standard continuous-time state-space form $\dot{\mathbf{x}} = A\mathbf{x} + B\mathbf{u}$, with

$$A = \begin{bmatrix} -\dfrac{R_1 + R_3}{L_1} & \dfrac{R_3}{L_1} & -\dfrac{1}{L_1} \\ \dfrac{R_3}{L_2} & -\dfrac{R_2 + R_3}{L_2} & \dfrac{1}{L_2} \\ \dfrac{1}{C} & -\dfrac{1}{C} & 0 \end{bmatrix}, \qquad B = \begin{bmatrix} \dfrac{1}{L_1} & 0 \\ 0 & -\dfrac{1}{L_2} \\ 0 & 0 \end{bmatrix}. \tag{28}$$

**Discretization and time stepping.**

Because the simulation operates on a fixed time step $\Delta t$, the continuous-time system is converted to its discrete-time equivalent $(A_d, B_d)$ using a zero-order-hold transformation. The state is then propagated forward in time by the recurrence

$$\mathbf{x}_{k+1} = A_d \mathbf{x}_k + B_d \mathbf{u}_k, \tag{29}$$

starting from zero initial conditions, where $\mathbf{u}_k = [V_s[k], V_g[k]]^T$.

**Derived quantities.**

Once the state trajectory is obtained, the capacitor current and the inductor voltages are recovered algebraically from the state variables:

$$I_C = I_{L_1} - I_{L_2}, \tag{30}$$

$$V_{L_1} = V_s - V_C - R_3 I_C - R_1 I_{L_1}, \tag{31}$$

$$V_{L_2} = V_C + R_3 I_C - V_g - R_2 I_{L_2}. \tag{32}$$

Finally, the solution is verified against Kirchhoff's current and voltage laws; if any consistency check exceeds tolerance, a warning is issued.

### 3.4.4 Capacitor_total_power_losses

This function computes the total power dissipated in the LCL-filter capacitor by summing the loss contributions across all relevant harmonics. The capacitor current is not purely sinusoidal: it contains the fundamental and low-order grid harmonics together with the high-frequency components concentrated around the switching frequency and its multiples. Each harmonic produces both a dielectric and an ohmic loss, and the two mechanisms dominate in different parts of the spectrum.

**Harmonic selection.**

The loss is evaluated only at the harmonic orders that carry significant capacitor current. These comprise the dominant low-order grid harmonics $(1, 5, 7, 11, 13, 17, 19)$ and two sidebands centred on the first and second multiples of the frequency-modulation ratio $m_f = f_{sw}/f$, namely $\{m_f - 2, \ldots, m_f + 2\}$ and $\{2m_f - 2, \ldots, 2m_f + 2\}$. The absolute frequency of harmonic $i$ is $f_i = i\, f$.

**Dielectric losses.**

The dielectric loss of the film capacitor is represented through its dissipation factor (loss tangent) $\tan\delta_0$, following the manufacturer loss model [2]. The dissipation factor is defined as the ratio of the equivalent dielectric series resistance $R_d$ to the capacitive reactance:

$$\tan\delta_0 = \frac{R_d}{X_C} = R_d\, \omega_i C \quad \Longrightarrow \quad R_d = \frac{\tan\delta_0}{2\pi f_i C}, \tag{33}$$

where $\omega_i = 2\pi f_i$. The dielectric power dissipated at harmonic $i$, carrying an RMS current $I(f_i)$, is therefore

$$P_D(f_i) = I(f_i)^2\, R_d = I(f_i)^2\, \frac{\tan\delta_0}{2\pi f_i C}. \tag{34}$$

This contribution decreases with frequency and dominates at the fundamental and low-order harmonics.

**Ohmic losses.**

The ohmic loss follows directly from Joule heating in the capacitor series resistance $R_s$:

$$P_R(f_i) = I(f_i)^2\, R_s. \tag{35}$$

Unlike the dielectric term, this loss is frequency-independent and therefore dominates in the switching-frequency sidebands, where the dielectric contribution has decayed.

**Total loss.**

The total capacitor dissipation for each mission-profile sample is obtained by summing both mechanisms over all considered harmonics:

$$P_{total,C} = \sum_i \left[P_D(f_i) + P_R(f_i)\right] = \sum_i I(f_i)^2 \left(\frac{\tan\delta_0}{2\pi f_i C} + R_s\right). \tag{36}$$

#### 3.4.5 calculate_tan_delta_0

This function extracts the frequency-independent dielectric loss tangent $\tan\delta_0$ of the polypropylene film from the total dissipation factor specified in the capacitor datasheet. The manufacturer specifies the total dissipation factor at a single measurement frequency, but this value combines two distinct loss mechanisms: a frequency-independent dielectric contribution and a frequency-dependent resistive contribution arising from the series resistance. According to the manufacturer loss model [2], the total dissipation factor is

$$\tan\delta(f) = \tan\delta_0 + R_s\,\omega\,C, \tag{37}$$

where $\omega = 2\pi f$, $R_s$ is the capacitor series resistance, and $C$ the capacitance. Rearranging to isolate the dielectric term gives

$$\tan\delta_0 = \tan\delta(f) - R_s\,\omega\,C, \tag{38}$$

evaluated at the datasheet measurement frequency (here 100 Hz). The function raises an error if the resistive term exceeds the measured dissipation factor, which would yield a non-physical negative $\tan\delta_0$ and indicate inconsistent input values.

#### 3.4.6 calculate_capacitor_thermal_resistance

This function determines the thermal resistance $R_{th,C}$ from the capacitor to the ambient, used to convert the capacitor loss into a temperature rise. Two methods are supported. In the `"user"` method, $R_{th,C}$ is supplied directly from a datasheet or measurement. In the `"surface_area"` method, it is estimated from the cylindrical can geometry and the convective heat-transfer coefficient [3].

The exposed surface of the capacitor is modeled as a cylinder, comprising the lateral side area and the two circular end caps:

$$A_{surface} = \underbrace{\pi D H}_{\text{lateral}} + \underbrace{2\,\pi\left(\frac{D}{2}\right)^2}_{\text{end caps}}, \tag{39}$$

where $D$ is the can diameter and $H$ its height. The thermal resistance is then the inverse of the convective conductance over this surface:

$$R_{th,C} = \frac{1}{h\,A_{surface}}, \tag{40}$$

with $h$ the convective heat-transfer coefficient, whose value depends on the cooling mode (around $10\ \mathrm{W/(m^2K)}$ for natural convection in still air, higher for forced cooling).

#### 3.4.7 Capacitor_hotspot_temperature

This function estimates the capacitor hotspot temperature from the ambient temperature and the total internal power dissipation, using a steady-state lumped thermal model. The capacitor is represented by a single thermal resistance $R_{th,C}$ between the internal hotspot and the surrounding ambient, so that the temperature rise above ambient is proportional to the dissipated power:

$$T_C = T_{amb} + P_{total,C}\,R_{th,C}, \tag{41}$$

where $T_{amb}$ is the ambient temperature, $P_{total,C}$ the total capacitor loss, and $R_{th,C}$ the hotspot-to-ambient thermal resistance.

The model assumes steady-state operation, meaning the thermal capacitance of the capacitor is neglected and the hotspot temperature is taken to follow the instantaneous loss directly. This is justified for mission-profile analysis, where each operating point is held long enough relative to the capacitor thermal time constant for the temperature to settle. A single-node (rather than a distributed or Foster/Cauer) representation is used because capacitor datasheets typically provide only a single thermal resistance and no thermal time constants.

### 3.4.8 calculate_core_surface_area

This function computes the total exposed surface area of a rectangular tape-wound inductor core, which is subsequently used to estimate the core-to-ambient thermal resistance through natural convection. The core has a rectangular picture-frame cross-section of depth $D$, with an outer rectangular boundary ($A \times B$) and an inner rectangular window ($F \times G$) through which the winding passes. The total surface area is the sum of three contributions:

$$A_{surface} = \underbrace{2(A+B)\,D}_{\text{outer side faces}} + \underbrace{2(F+G)\,D}_{\text{inner window faces}} + \underbrace{2\,(AB-FG)}_{\text{front and back faces}}, \tag{42}$$

where $A$ and $B$ are the overall outer width and height, $F$ and $G$ are the window width and height, and $D$ is the core depth. The first term accounts for the four outer lateral faces, the second for the four inner faces lining the window opening, and the third for the two end faces, whose area is the outer rectangle minus the window cutout.

### 3.4.9 calculate_Ae

This function determines the effective magnetic cross-sectional area $A_e$ of the inductor core, which is the area through which the magnetic flux effectively flows. Two methods are supported. In the `"user"` method, $A_e$ is taken directly from the manufacturer datasheet, where the stacking factor has already been accounted for. In the `"geometry"` method, $A_e$ is computed from the core cross-sectional dimensions and the stacking factor [4]:

$$A_e = k_f\,D\,E, \tag{43}$$

where $D$ is the core depth, $E$ is the build (thickness), and $k_f$ is the stacking factor. The stacking factor ($k_f < 1$) accounts for the non-magnetic gaps between the wound lamination layers, so that $A_e$ represents only the magnetic material rather than the full geometric cross-section. The geometry method is used when scaling an existing core or estimating a custom core size.

### 3.4.10 calculate_le

This function determines the effective magnetic path length $l_e$ of the inductor core, that is, the average distance the magnetic flux travels around the closed core loop. Two methods are supported. In the `"user"` method, $l_e$ is taken directly from the manufacturer datasheet, which is preferred when available because it accounts for the corner rounding of the wound core. In the `"geometry"` method, $l_e$ is estimated as the perimeter of the centreline rectangle passing through the middle of the core material [4].
The flux travels not along the outer edge nor the inner window edge, but through the midpoint of each leg. With the horizontal and vertical leg widths defined as

$$\text{leg}_w = \frac{A-F}{2}, \qquad \text{leg}_h = \frac{B-G}{2}, \tag{44}$$

the centreline perimeter is

$$l_e = 2\,(F + \mathrm{leg}_w) + 2\,(G + \mathrm{leg}_h) = (A + F) + (B + G), \tag{45}$$

where $A$ and $B$ are the outer width and height of the core and $F$ and $G$ are the window width and height. This geometric estimate carries a small residual error (on the order of 10%) relative to the datasheet value, arising from the corner rounding of the wound core, which slightly shortens the actual flux path.

### 3.4.11 calculate_turns

This function computes the minimum number of winding turns $N$ required to realize the target inductance without driving the core into saturation. Starting from the definition of inductance in terms of the flux linkage [4],

$$L = \frac{N\,\Phi}{I} = \frac{N\,B\,A_e}{I}, \tag{46}$$

the flux density at a given current is

$$B = \frac{L\,I}{N\,A_e}. \tag{47}$$

At the peak current $I_{peak}$, the flux density must remain below the maximum allowable value $B_{max}$ to avoid saturation:

$$B_{peak} = \frac{L\,I_{peak}}{N\,A_e} \leq B_{max}. \tag{48}$$

Solving for the number of turns gives the lower bound

$$N \geq \frac{L\,I_{peak}}{B_{max}\,A_e}. \tag{49}$$

The result is rounded up to the nearest integer, since the number of turns must be a whole number and rounding down would raise the peak flux density above $B_{max}$, risking saturation:

$$N = \left\lceil \frac{L\,I_{peak}}{B_{max}\,A_e} \right\rceil. \tag{50}$$

### 3.4.12 calculate_B_peak

This function computes the peak magnetic flux density $B_{peak}$ in the inductor core at peak current, accounting for both the core material and the air gap. Applying Ampere's law around the closed magnetic loop, the magnetomotive force $NI$ is balanced by the reluctance drops across the core and the air gap [4]:

$$NI = \frac{B\,l_e}{\mu_0\mu_r} + \frac{B\,l_g}{\mu_0} = \frac{B}{\mu_0}\left(\frac{l_e}{\mu_r} + l_g\right), \tag{51}$$

where $l_e$ is the magnetic path length, $l_g$ the air-gap length, $\mu_r$ the relative permeability of the core material, and $\mu_0$ the permeability of free space. Solving for the flux density at the peak current $I_{peak}$ gives

$$B_{peak} = \frac{\mu_0\,N\,I_{peak}}{l_g + \dfrac{l_e}{\mu_r}}. \tag{52}$$

The air-gap term $l_g$ dominates the denominator because the gap, although physically short, has a reluctance far larger than that of the high-permeability core (where $l_e/\mu_r$ is small), so the gap is what primarily sets the peak flux density.

### 3.4.13 calculate_minimum_required_wire_area

This function computes the minimum copper cross-sectional area required for the winding to carry the rated current without exceeding the maximum allowable current density. The current density $J$ in a conductor is the current per unit cross-sectional area; limiting it to a maximum value $J_{max}$ keeps the winding losses and the associated temperature rise within acceptable bounds. The minimum required copper area is therefore

$$A_{wire} = \frac{I_{RMS}}{J_{max}}, \tag{53}$$

where $I_{RMS}$ is the rated RMS current. The equivalent minimum wire diameter, assuming a circular cross-section, follows from $A = \pi d^2/4$:

$$d_{wire} = \sqrt{\frac{4\,A_{wire}}{\pi}}. \tag{54}$$

### 3.4.14 calculate_Rdc

This function computes the DC resistance of the inductor winding. The total length of copper is the number of turns $N$ multiplied by the mean length of one turn $l_{turn}$, and the resistance of a conductor follows from the standard resistance–resistivity relation [4]:

$$R_{dc} = \frac{\rho\,N\,l_{turn}}{A_{wire}}, \tag{55}$$

where $\rho$ is the copper resistivity, $A_{wire}$ is the conductor cross-sectional area, and $N\,l_{turn}$ is the total winding length. Skin and proximity effects are neglected, so the AC resistance is taken equal to the DC resistance ($R_{ac} \approx R_{dc}$); this is a simplification, since at the switching-frequency harmonics the effective resistance would in practice be somewhat higher.

### 3.4.15 calculate_inductor_core_losses

This function computes the core (magnetic) loss of the inductor by applying the Steinmetz equation to every harmonic of the inductor current [5]. Because the current is non-sinusoidal, the core loss is evaluated harmonic by harmonic and summed, rather than from the fundamental alone.

**Step 1 – Peak flux density per harmonic.**

For each harmonic $j$, the peak flux density in the gapped core is obtained from Ampere's law (as in Section 3.4.12), using the peak current of that harmonic:

$$B_j = \frac{\mu_0\,N\,I_{peak,j}}{l_g + \dfrac{l_e}{\mu_r}}. \tag{56}$$

**Step 2 – Core loss per harmonic.**

The volumetric core loss at each harmonic is given by the Steinmetz equation, scaled by the effective core volume $V_e$:

$$P_{c,j} = k\,f_j^{\alpha}\,B_j^{\beta}\,V_e, \tag{57}$$

where $f_j$ is the harmonic frequency, $k$ is the Steinmetz loss coefficient, and $\alpha$ and $\beta$ are the Steinmetz frequency and flux-density exponents, respectively. These coefficients are material-specific and are fitted to the core manufacturer loss data.

**Step 3 – Total core loss.**

The total core loss for each mission-profile sample is the sum of the contributions of all considered harmonics:

$$P_c = \sum_j P_{c,j} = \sum_j k\, f_j^{\alpha}\, B_j^{\beta}\, V_e. \tag{58}$$

### 3.4.16 calculate_winding_losses

This function computes the copper (winding) loss of the inductor for each mission-profile sample. The loss is the ohmic dissipation in the winding resistance carrying the full inductor current. The RMS value of the instantaneous inductor current is first computed over each one-second window,

$$I_{RMS} = \sqrt{\frac{1}{n}\sum_{i=1}^{n} I_L[i]^2}, \tag{59}$$

where $n$ is the number of samples per second, and the winding loss then follows from Joule heating:

$$P_w = R_{dc}\, I_{RMS}^2. \tag{60}$$

Because the DC winding resistance $R_{dc}$ is used (Section 3.4.14), the full current waveform, including its switching-frequency harmonics, is treated as if it experiences the same resistance; skin and proximity effects are neglected for model simplicity.

### 3.4.17 calculate_inductor_thermal_resistance

This function determines the thermal resistance $R_{th}$ from the inductor to the ambient, which converts the total inductor loss into a temperature rise. Three methods are supported. In the `"user"` method, $R_{th}$ is supplied directly from a datasheet or measurement. The remaining two methods estimate it from the core geometry.

**Surface-area method.**

This method models the inductor as a single thermal node losing heat to the ambient by convection over its exposed surface. The thermal resistance is the inverse of the convective conductance [3]:

$$R_{th} = \frac{1}{h\, A_{surface}}, \tag{61}$$

where $A_{surface}$ is the total exposed core surface area (Section 3.4.8) and $h$ is the convective heat-transfer coefficient. A single lumped $R_{th}$ is assumed to represent the combined thermal path of the entire inductor assembly to ambient, with the core and winding taken to be at the same temperature [5]. The heat-transfer coefficient depends on the cooling mode, ranging from about 10 $\mathrm{W/(m^2K)}$ for natural convection in still air to several hundred for forced-air or liquid cooling.

**Empirical method.**

Alternatively, $R_{th}$ is estimated from the effective core volume using the empirical relation for naturally cooled magnetic cores [4]:

$$R_{th} = \frac{14.5}{V_e^{0.37}}, \tag{62}$$

where $V_e$ is the effective core volume expressed in $\mathrm{cm^3}$ and $R_{th}$ is obtained in K/W.

#### 3.4.18 calculate_inductor_temperature

This function estimates the inductor temperature from the ambient temperature and the total inductor loss using a single-node steady-state thermal model. The combined core and winding losses drive a temperature rise above ambient proportional to the lumped thermal resistance [5]:

$$T_{inductor} = T_{amb} + R_{th} P_{total}, \qquad P_{total} = P_c + P_w, \tag{63}$$

where $T_{amb}$ is the ambient temperature, $R_{th}$ the inductor-to-ambient thermal resistance (Section 3.4.17), and $P_{total}$ the sum of the core loss $P_c$ and winding loss $P_w$. As in the capacitor thermal model (Section 3.4.7), the thermal capacitance is neglected, so the temperature follows the instantaneous loss directly; this is justified for mission-profile analysis where each operating point persists long relative to the inductor thermal time constant.

## 3.5 Plotting_function

This module provides the post-processing and visualization utilities for the LCL-filter lifetime framework. It is implemented as a single class, `Plotting_functions_class`, containing three static methods that each consume one of the simulation output dataframes and generate the corresponding set of figures. The first method visualizes the RMS mission-profile quantities, the second the instantaneous one-cycle waveforms and the third the per-component electrical, loss, thermal, and lifetime results for the two inductors and the capacitor. All figures are written as `.png` files to the supplied output directory. Table 14 summarizes the three methods and Table 15 lists the individual figures produced.

Table 14: Plotting methods.

| **Method** | **Description** |
|---|---|
| `plot_df_1_power_flow_RMS` | Plots the RMS mission-profile quantities (apparent, active and reactive power; grid voltage and current; power factor). |
| `plot_df_2_power_flow_inst` | Plots the instantaneous waveforms over the last fundamental cycle (reference signals, switched inverter voltage, and the per-component voltage/current waveforms) together with the THD bar chart. |
| `plot_df_components` | Plots the per-component RMS quantities, losses, temperatures and lifetimes for the capacitor $C$ and the two inductors $L_1$ and $L_2$. |

Table 15: Figures generated by the plotting module.

| **Figure** | **Description** |
|---|---|
| Fig. 1 | Apparent, active and reactive power versus time. |
| Fig. 2 | Grid-side RMS voltage $V_g$ and current $I_g$ versus time. |
| Fig. 3 | Power factor versus time. |
| Fig. 4 | Instantaneous power factor and phase angle $\phi$. |
| Fig. 5 | Reference grid current $I_{g,ref}$ and reference inverter voltage $V_{s,ref}$. |
| Fig. 6 | Switched inverter output voltage $V_s$. |
| Fig. 7 | Inverter-side inductor voltage $V_{L_1}$ and current $I_{L_1}$. |
| Fig. 8 | Capacitor voltage $V_C$ and current $I_C$. |
| Fig. 9 | Grid-side inductor voltage $V_{L_2}$ and current $I_{L_2}$. |
| Fig. 10 | Total harmonic distortion of $I_{L_2}$ (bar chart). |
| Fig. 11 | Grid-side current $I_{L_2}$ compared against the reference current $I_{g,ref}$. |
| Fig. 12 | Capacitor RMS voltage and current versus time. |
| Fig. 13 | Inverter-side inductor RMS voltage and current versus time. |
| Fig. 14 | Grid-side inductor RMS voltage and current versus time. |
| Fig. 15 | Total power loss in $C$, $L_1$ and $L_2$ versus time. |
| Fig. 16 | Core and winding losses of the two inductors versus time. |
| Fig. 17 | Estimated core/hotspot temperatures of $C$, $L_1$ and $L_2$ versus time. |
| Fig. 18 | Equivalent lifetime of each component (bar chart). |
| Fig. 19 | Lifetime consumed over the mission profile (bar chart). |
| Fig. 20 | Average operating temperature of each component (bar chart). |

# 4 Benchmarking

The lifetime predictions produced by the framework ultimately rest on the accuracy of the underlying electrical model: the inductor and capacitor losses, temperatures, and lifetimes are all derived from the voltage and current waveforms computed by the LCL-filter solver. Before these downstream predictions can be trusted, the electrical model itself must be shown to reproduce the behaviour of the physical filter. This section therefore benchmarks the electrical quantities computed by the Python model against an independent, industry-standard circuit simulator.

**Reference simulator.**

The benchmark reference is PLECS, a widely used power-electronics simulation tool that solves the switched circuit using an event-driven solver, placing each switching transition at its exact crossing time. PLECS is chosen because it is independent of the present implementation, is established as a trusted reference in power-electronics research, and resolves the switching behaviour at a fidelity higher than the fixed-step Python model. It therefore serves as an appropriate ground truth against which the Python results are compared. The PLECS benchmarking model is shown in Fig. 1.

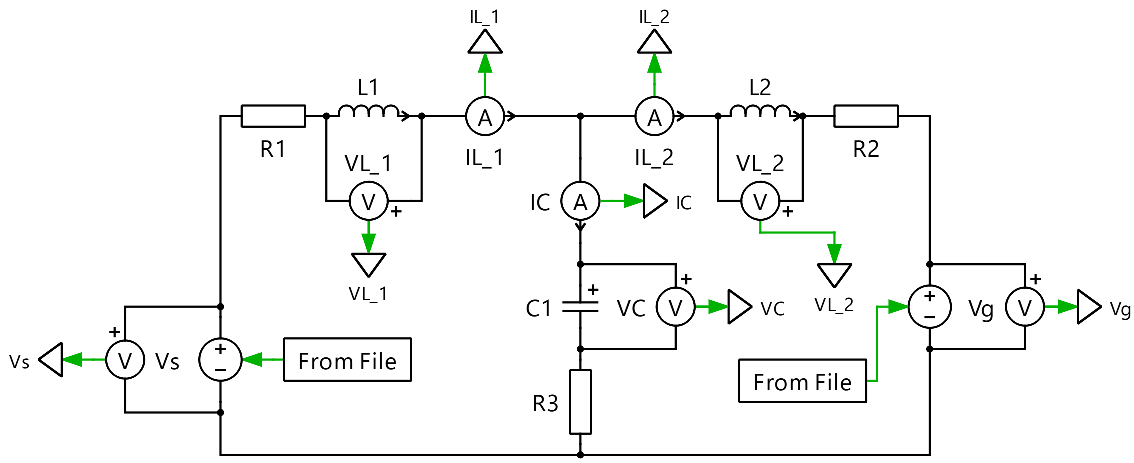

Figure 1: PLECS benchmarking model of the LCL filter.

**Methodology.**

To ensure that both tools solve an identical electrical problem, the comparison is performed in an open-loop manner. The reference inverter voltage $V_{s,ref}$ and switched inverter voltage $V_s$ generated by the Python model are exported and used to drive an identical LCL-filter network in PLECS, built with the same component values $L_1$, $L_2$, $C$ and the parasitic resistances $R_1$, $R_2$, $R_3$. Both models are run to steady state, and the final fundamental cycle is used for comparison. The same set of quantities is then extracted from each tool: the RMS values of all filter voltages and currents ($V_{L_1}$, $I_{L_1}$, $V_C$, $I_C$, $V_{L_2}$, $I_{L_2}$, $V_g$, $V_s$), the power factor, the total harmonic distortion of the grid-side current $I_{L_2}$, and the full grid-current waveform. Agreement is quantified by the relative percentage error for the scalar quantities and by the normalized root-mean-square error (NRMSE) for the grid-current waveform.

**Operating points.**

The comparison is carried out over eight operating points spanning the inverter operating envelope. A power sweep at unity power factor (1.00, 0.75, 0.50 and 0.25 MVA) verifies the model across the loading range, and a power-factor sweep at rated power ($pf = 0.866$ and 0.5, both lagging and leading) verifies it under reactive operation. Together these probe both the magnitude and the phase behaviour of the filter.

**Scope of the benchmark.**

The benchmarking is deliberately restricted to the *electrical* domain. The thermal and lifetime models are not benchmarked against PLECS, for two reasons. First, the loss, thermal, and lifetime models are analytical formulations driven directly by the electrical quantities; once the electrical waveforms are validated, the remaining stages are deterministic algebraic mappings rather than independent simulations, so validating the electrical input is what establishes confidence in the chain. Second, PLECS does not provide an equivalent built-in inductor-insulation or capacitor wear-out lifetime model, so a like-for-like comparison of temperature or lifetime is not available. The electrical benchmark therefore validates the foundation on which the thermal and lifetime predictions are built.

## 4.1 Results and discussion

Table 16 reports the comparison between the Python model and PLECS across all eight operating points. The scalar quantities are expressed as the relative percentage error, and the grid-side current waveform is compared using the NRMSE.

Table 16: Benchmarking of the Python model against PLECS (PLECS as reference).

| Quantity | Power sweep (pf = 1.0) | | | | Power-factor sweep (1.0 MVA) | | | |
|---|---|---|---|---|---|---|---|---|
| | 1.00 MVA | 0.75 MVA | 0.50 MVA | 0.25 MVA | 0.866 lag | 0.5 lag | 0.866 lead | 0.5 lead |
| $V_{L_1}$ RMS [% err] | 4.82 | 4.81 | 4.94 | 4.97 | 4.84 | 4.78 | 4.81 | 4.97 |
| $I_{L_1}$ RMS [% err] | 0.00 | 0.01 | 0.01 | 0.06 | 0.00 | 0.07 | 0.07 | 0.14 |
| $I_C$ RMS [% err] | 2.05 | 2.13 | 2.22 | 2.29 | 2.05 | 2.11 | 2.13 | 2.28 |
| $V_C$ RMS [% err] | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.01 |
| $V_{L_2}$ RMS [% err] | 3.21 | 2.77 | 1.97 | 0.61 | 2.68 | 1.32 | 2.86 | 1.48 |
| $I_{L_2}$ RMS [% err] | 0.00 | 0.00 | 0.00 | 0.01 | 0.00 | 0.08 | 0.07 | 0.15 |
| $pf$ [% err] | 0.00 | 0.00 | 0.00 | 0.01 | 0.05 | 0.19 | 0.08 | 0.16 |
| THD [% err] | 0.40 | 0.03 | 0.33 | 0.52 | 0.58 | 0.99 | 0.78 | 0.73 |
| $I_{L_2}$ [NRMSE] | 0.09 | 0.11 | 0.16 | 0.31 | 1.30 | 2.32 | 1.44 | 2.45 |

**Quantities that agree closely.**

The grid-side current $I_{L_2}$, the inverter-side current $I_{L_1}$, the capacitor voltage $V_C$ and the power factor all match PLECS to within a small fraction of a percent across every operating point, with most errors below 0.1%. These are the smooth, filtered quantities of the LCL filter, dominated by the fundamental component. Their close agreement confirms that the state-space filter solver, the phasor reference computation, and the power-factor handling are all implemented correctly. Since these are precisely the quantities on which the downstream loss, thermal and lifetime calculations depend, their accuracy is the central requirement of the benchmark, and it is met.

**Quantities with a larger error.**

Three quantities show a larger discrepancy: $V_{L_1}$ (consistently about 4.8%), $I_C$ (about 2.1%), and $V_{L_2}$ (between 0.6% and 3.2%). These are not smooth fundamental quantities but are dominated almost entirely by the high-frequency switching ripple, as is evident from Fig. 2: the inductor voltages and the capacitor current consist almost entirely of dense, rapidly varying switching content with little or no smooth fundamental component.The RMS value of such a signal is determined by the exact placement and width of every switching edge.

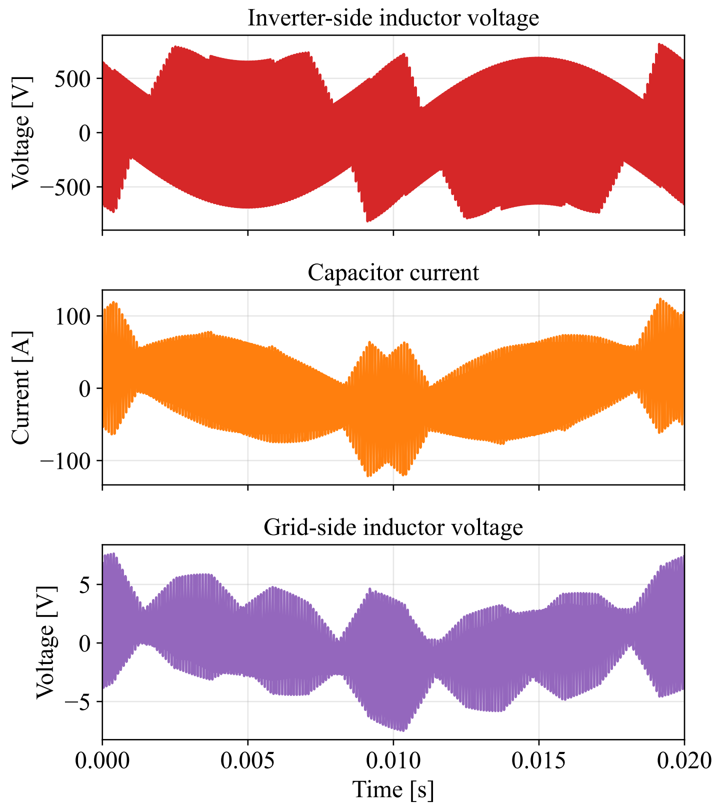


Figure 2: Switching-dominated waveforms over one fundamental cycle.

The Python model and PLECS represent these switching edges differently. The Python model renders the pulses onto a fixed time grid using volt-second-preserving averaging (Section 3.4.1), whereas PLECS resolves each edge exactly with an event-driven solver. The two therefore see slightly different switched waveforms at the microsecond level, even though their local averages agree. Because $V_{L_1}$, $I_C$ and $V_{L_2}$ are built almost entirely from this switching content, a small difference in edge representation maps directly into a few-percent RMS difference. Expecting near-perfect agreement on quantities that are essentially pure ripple is therefore unrealistic. The discrepancy can be reduced by increasing the simulation resolution, which places the edges more finely, but only at the cost of higher computation time and memory usage; the present resolution is a deliberate compromise between accuracy and cost.

**Grid-current waveform agreement.**

The ultimate quantity of interest is the grid-side current $I_{L_2}$, since it is the filtered output delivered to the grid and the dominant driver of the grid-side inductor losses. The NRMSE between the Python and PLECS $I_{L_2}$ waveforms remains low across all operating points, ranging from 0.09% at full load to a few percent at low power factor. The close overlay of the two waveforms is shown in Fig. 3. The slightly higher NRMSE at low power factor and low power reflects the smaller fundamental current there, which makes the residual switching mismatch a larger fraction of the signal; the absolute error remains small.

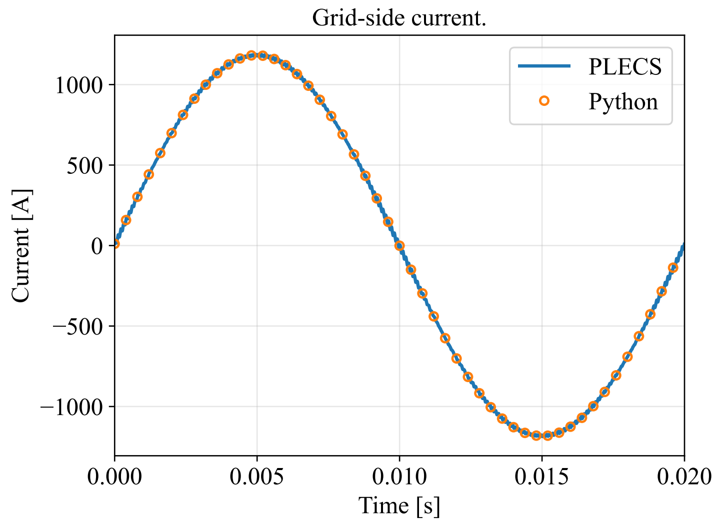


Figure 3: Grid-side current from PLECS and the Python model over one fundamental cycle.

**Conclusion.**

The benchmark is considered successful. The filtered quantities that govern the loss, thermal and lifetime models, namely $I_{L_1}$, $I_{L_2}$, $V_C$ and the power factor, agree with PLECS to well within 0.2%, and the grid-side current waveform matches with a low NRMSE. The larger errors are confined to the purely switching-dominated quantities ($V_{L_1}$, $I_C$, $V_{L_2}$) and are fully explained by the difference in switching-edge representation between a fixed-step and an event-driven solver, rather than by any error in the electrical model. The electrical foundation of the framework is therefore validated.

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